

Referee Report (Contribution Focus) for “Regulating Competing Payment Networks”

1 Summary of the Paper

This paper develops and estimates a structural model of payment network competition to evaluate how fee caps, network competition, and dual-routing requirements affect consumer welfare. The author models a three-stage game in which profit-maximizing networks set merchant fees and consumer rewards, consumers and merchants make adoption decisions, and consumers make consumption choices across merchants. The model features consumer and merchant multi-homing, merchant pricing under price coherence, and network competition.

The paper uses the 2011 Durbin Amendment as a quasi-experiment to identify consumers’ sensitivity to rewards, event-study evidence from grocery chains’ credit card acceptance decisions to identify merchants’ sales gains from card acceptance, and second-choice surveys to estimate substitution patterns. The author finds that consumers are ten times more sensitive to rewards than merchants are to fees. This asymmetry creates a “competitive bottleneck” in which networks compete for consumers through high rewards funded by high merchant fees. The main counterfactual results show that capping credit and debit card fees at the cost of cash would increase annual welfare by \$30 billion, while a merger to monopoly would increase welfare by \$16 billion by reducing rewards competition.

2 Contribution Assessment

The paper’s contribution is to develop and estimate the first quantitative model of payment network competition that combines consumer multi-homing, merchant heterogeneity, and merchant competition to evaluate the welfare effects of fee regulation and network competition.

Importance: The paper addresses a question of substantial policy relevance. Merchant fees total over \$120 billion annually in the U.S., and payment market regulation is actively debated (the 2022 Credit Card Competition Act, the 2024 DOJ lawsuit against Visa). Understanding whether fee caps versus competition better addresses market failures has clear policy implications.

Magnitude of advance: The paper makes a meaningful advance over existing work. While Rochet

and Tirole (2003, 2011), Armstrong (2006), and Edelman and Wright (2015) provide theoretical foundations, and Huynh et al. (2022) estimate a structural two-sided model, this is the first paper to endogenize network pricing alongside consumer and merchant decisions. This allows the author to trace how changes in regulation or competition affect equilibrium fees, rewards, market shares, and welfare—something prior work could not do.

Novelty: The claimed novelty is accurate. The paper correctly identifies that prior structural work (Huynh et al. 2022) did not model network competition, that theoretical models typically assume either single-homing or full multi-homing, and that the combination of these features is essential for credible policy predictions. The finding that competition reduces welfare is a genuinely surprising result that emerges from the model rather than being assumed.

Broad relevance: The paper has broad relevance beyond payment economists. The result that competition can exacerbate market failures under price coherence speaks to antitrust economists and policymakers. The finding that fee regulation is progressive connects to public finance. The competitive bottleneck mechanism applies to other two-sided platforms.

3 Literature Positioning

The paper demonstrates strong command of the relevant literature. The author correctly positions the work relative to the foundational theory (Rochet and Tirole 2003, 2011; Armstrong 2006; Wright 2004), the price coherence literature (Edelman and Wright 2015), and the empirical IO literature on two-sided markets (Rysman 2004, Lee 2013, Song 2021).

The paper appropriately cites Huynh et al. (2022) as the closest structural antecedent and clearly explains the gap: Huynh et al. model consumer and merchant adoption but not network competition. The paper also engages with recent work on the Durbin Amendment (Mukharlyamov and Sarin 2025) and uses it as both an identification strategy and a robustness benchmark.

Two small gaps in the literature discussion:

- The paper could engage more with the empirical work on pass-through of interchange fee reductions in Australia and Europe (e.g., Gans 2007, Valverde et al. 2016), particularly given the importance of pass-through assumptions for welfare results.
- The paper does not cite Koulayev et al. (2016), who also estimate a structural model separating adoption from usage decisions using U.S. payment data. A brief discussion of how their approach differs would strengthen the methods section.

4 Deal Breakers

None identified from contribution perspective.

The paper makes a clear, novel contribution that advances both the academic literature and policy understanding. The structural estimation of network competition in payment markets is genuinely new, and the main finding—that competition can reduce welfare under price coherence—is an important theoretical prediction that prior work identified but could not quantify.

5 Other Issues

- The introduction is well-written but could more explicitly state why a structural model is necessary. The author implicitly argues that reduced-form estimates cannot evaluate counterfactuals involving changes in network conduct, but this could be stated more directly.
- The \$30 billion welfare gain from fee caps is a headline result. The paper would benefit from more explicit benchmarking of this magnitude (beyond the comparison to the CARD Act). For instance: What fraction of total merchant fees does this represent? How does it compare to the deadweight loss from other retail market frictions?
- The “credit aversion” mechanism is central to the welfare results but receives relatively brief discussion. Additional evidence on what generates credit aversion (beyond the survey evidence in the appendix) would strengthen the interpretation.
- The merger counterfactual is provocative and clarifies the mechanism, but the policy implications are unclear. The paper should be explicit that this is a thought experiment to illustrate the direction of competition effects, not a policy recommendation.
- The paper focuses on U.S. markets but draws lessons for regulation that has been implemented in Europe and Australia. A brief discussion of why U.S.-estimated parameters may or may not apply to other markets would add nuance.